

Theory Selection and Divergence Kernels Across Domains

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Abstract

Given a fixed ontology, multiple consistent theories may coexist over it, and the truth value of a proposition about a fixed object may depend on which theory is active. We define the divergence kernel of two theories T_1 and T_2 as the set $\Delta(T_1, T_2) = \{\varphi \mid T_1 \models \varphi \wedge T_2 \not\models \varphi\}$ — the minimal collection of predicates on which the theories disagree. We demonstrate computationally that across four domains — set theory, international law, tonal music, and chess — the divergence kernel is small, precisely identifiable, and corresponds to a recognizable domain concept: the Continuum Hypothesis in set theory, the imminent-attack predicate in international law, parallel-fifth tolerance in music theory, and proof scaffolding in chess. All four domains are formalized on the same verification platform (Kleis + Z3), enabling direct structural comparison. We distinguish hard constraints (chess: axioms are inviolable) from soft constraints (music: axioms are selectively violable), and show that theory selection — the choice of which theory to apply to a given object — is an extra-formal act: the mathematician’s foundational commitment, the jurist’s doctrinal stance, the composer’s aesthetic judgment. The formal system can compute the divergence kernel but cannot determine which side of it to inhabit. We argue that this boundary — computable disagreement, uncomputeable selection — is a structural invariant of reasoning under constraints.

Keywords: divergence kernel, theory selection, formal verification, model theory, SMT solving, soft constraints, hard constraints, Kleis

1 Introduction

When two theories share an ontology but disagree on specific propositions, where exactly does the disagreement live? And can a machine identify it?

Consider a concrete scenario: two legal doctrines share the same vocabulary (states, acts, norms) and the same baseline law (force is prohibited unless an exception applies). They disagree on a single point: what counts as a valid exception. Given a specific act — say, a military strike where no armed attack has occurred — one doctrine deems it prohibited; the other, under certain evidence conditions, may deem it permitted. The entire disagreement collapses to a single predicate: **is the imminent-attack condition satisfiable?**

This localization of disagreement is not unique to law. In set theory, the entire disagreement between models that satisfy the Continuum Hypothesis and those that do not collapses to a single statement. In music, the difference between Bach-style and Beethoven-style composition can be

characterized by which voice-leading axioms each satisfies. In chess, competing proof strategies for a puzzle differ in which intermediate lemmas (scaffolding) they supply.

In each case, two theories T_1 and T_2 share a vocabulary and agree on most propositions, but diverge on a small, identifiable set. We call this set the **divergence kernel**:

$$\Delta(T_1, T_2) = \{\varphi \mid T_1 \models \varphi \text{ and } T_2 \not\models \varphi\}$$

The divergence kernel is computable: load both theories into an SMT solver, evaluate the same propositions under each, and collect the disagreements. What is **not** computable is which theory to select. That choice depends on extra-formal considerations: foundational commitments in mathematics, doctrinal stances in law, aesthetic judgments in music, strategic insight in chess.

This paper makes three contributions:

1. We define the divergence kernel formally and show that it is small across four implemented domains.
2. We distinguish **hard constraints** (where all axioms must hold) from **soft constraints** (where axioms may be selectively violated), and show that this distinction determines whether theory selection operates at the witness level or the theory level.
3. We demonstrate all four domains on the same verification platform (Kleis + Z3), enabling direct structural comparison and ruling out the possibility that the pattern is an artifact of domain-specific formalizations.

2 The Divergence Kernel

Let $\mathcal{L}$ be a first-order language (an ontology) and let T_1, T_2 be theories over $\mathcal{L}$ — sets of sentences closed under logical consequence. We define:

$$\Delta(T_1, T_2) = \{\varphi \in \text{Sent}(\mathcal{L}) \mid T_1 \models \varphi \text{ and } T_2 \not\models \varphi\}$$

The divergence kernel contains exactly those sentences on which the theories disagree. If $\Delta(T_1, T_2) = \emptyset$, the theories are **logically equivalent** over $\mathcal{L}$. If Δ is nonempty but small, the disagreement is **localized**.

Several properties follow immediately:

- **Asymmetry:** $\Delta(T_1, T_2) \neq \Delta(T_2, T_1)$ in general, since $T_1 \models \varphi$ and $T_2 \not\models \varphi$ does not imply $T_2 \models \varphi$ and $T_1 \not\models \varphi$ (the latter requires $T_2 \models \neg\varphi$, which is stronger than $T_2 \not\models \varphi$). For decidable theories, both conditions coincide.
- **Closure under consequence:** If $\varphi \in \Delta(T_1, T_2)$ and $T_1 \models \varphi \rightarrow \psi$ but $T_2 \not\models \psi$, then $\psi \in \Delta$ as well. The kernel propagates through the deductive closure.
- **Minimal generators:** In practice, a large Δ often has a small set of **generators** — sentences from which the rest of Δ follows. We call these the **pivot predicates**. Identifying the pivot predicate is the main computational result in each domain.

In an SMT setting, the divergence kernel is computed operationally: assert the shared axioms, then query each theory-specific axiom under both configurations. A proposition φ is in Δ if it is SAT under T_1 and UNSAT under T_2 (or vice versa). The solver does not enumerate the full kernel; it answers point queries. But by querying strategically, we can identify the pivot.

3 Four Domains, One Pattern

We now present four domains, each formalized in Kleis with Z3 verification, where the divergence kernel is small and identifiable. The domains are deliberately diverse — pure mathematics, international law, tonal music, and combinatorial games — to demonstrate that the pattern is structural, not domain-specific.

3.1 Set Theory: The Continuum Hypothesis

The ontology $\mathcal{L}$ consists of cardinal numbers with ordering ($<$), arithmetic ($+$, $\cdot$, exponentiation), and a distinguished constant $\mathfrak{c}$ for the cardinality of the continuum.

Theory T_1 (ZFC + CH): $\mathfrak{c} = \aleph_1$ — there is no cardinality between $\aleph_0$ and $\mathfrak{c}$.

Theory T_2 (ZFC + $\neg$ CH): $\mathfrak{c} > \aleph_1$ — such intermediate cardinalities exist.

Both theories share the same cardinal arithmetic: $\aleph_0 + \aleph_0 = \aleph_0$, $2^{\aleph_0} = \mathfrak{c}$, the power-set growth law, well-ordering. They agree on every statement about natural numbers, every statement about countable ordinals, every theorem of ordinary mathematics. They disagree on exactly one pivot predicate:

$$\Delta(T_1, T_2) = \{\mathfrak{c} = \aleph_1\}$$

In our Kleis formalization (documented in a companion paper), the CH is detected as independent by constructing explicit models on both sides of a projection map Π from set-theoretic models to their cardinal-arithmetic shadows. Two models that project identically — same ordering, same arithmetic — differ on CH. Z3 verifies this non-invariance in under 30 seconds.

The divergence kernel has a single generator. All disagreement flows from one predicate.

3.2 International Law: Doctrine Selection

The ontology $\mathcal{L}$ consists of states, military acts, norms (Permitted/Prohibited), and predicates for force, self-defense, necessity, proportionality, armed attack, and imminent attack.

Theory T_1 (Strict doctrine): Self-defense requires that an armed attack has already occurred.

Theory T_2 (Anticipatory doctrine): Self-defense is also lawful when an attack is **imminent**, provided the response is necessary and proportional.

Both theories share Article 2(4) of the UN Charter (force is prohibited unless authorized or in lawful self-defense), the definitions of necessity and proportionality, and the Security Council authorization mechanism. The only structural difference is one disjunct in the self-defense precondition:

$$T_2 : \text{lawful_self_defense}(a) \leftarrow \text{necessary}(a) \wedge \text{proportional}(a) \wedge (\text{armed_attack}(\dots) \vee \text{imminent_attack}(\dots))$$

$$T_1 : \text{lawful_self_defense}(a) \leftarrow \text{necessary}(a) \wedge \text{proportional}(a) \wedge \text{armed_attack}(\dots)$$

Given a case where no armed attack has occurred but evidence supports imminence, T_1 yields Prohibited and T_2 yields Permitted. The pivot predicate is:

$$\Delta(T_1, T_2) = \{\text{imminent_attack}(\text{target}, \text{attacker}) \text{ satisfiable}\}$$

In our Kleis formalization, both doctrines coexist in a single file via a **Doctrine** type parameter that isolates their axioms. Z3 verifies that under a **NoEvidence** profile, both doctrines converge (both yield Prohibited). Under an **EvidenceSupported** profile, they diverge. The entire disagreement between two competing interpretations of international law collapses to a single predicate about the satisfiability of an imminence claim.

The selector here is the jurist — or, more precisely, the state invoking the doctrine. The choice is political, not logical.

3.3 Music: Style as Theory Selection

The ontology $\mathcal{L}$ consists of pitches, durations, measures, voices, and scores. Axioms encode voice-leading rules: prohibition of parallel octaves and fifths, smooth bass motion, triadic arpeggiation, melody–harmony consonance, and harmonic-rhythm uniformity.

Theory T_1 (Strict tonal harmony): All seven axioms must hold. This approximates textbook species counterpoint.

Theory T_2 (Tonal cohesion): A subset — tonic opening, smooth bass motion, melody–harmony consonance, no parallel octaves — must hold. Parallel fifths, strict triadic arpeggiation, and harmonic-rhythm uniformity are not required.

Beethoven’s Moonlight Sonata (Op. 27 No. 2, measures 1–14), formalized as a typed AST in Kleis, satisfies T_2 but not T_1 . The divergence kernel contains three generators:

$$\Delta(T_1, T_2) \supseteq \{\text{parallel_fifths_prohibited}, \text{strict_triadic_arpeggiation}, \text{uniform_harmonic_rhythm}\}$$

The violations are localized: parallel fifths at measure 13, non-triadic arpeggiation at measure 4, mid-bar harmonic changes in 8 of 14 measures. Each violation is machine-identified by functional evaluation over the concrete AST.

Music differs from the other domains in a crucial way: the axioms are **soft constraints**. Beethoven does not ‘fail’ by violating parallel-fifths prohibition — he **chooses** to satisfy a different sub-theory. The composer selects not just which witness (which specific notes) but which theory (which axioms to obey). Style, in this framework, is literally theory selection.

The selector is the composer. The choice is aesthetic, not logical.

3.4 Chess: Hard Constraints and Proof Scaffolding

The ontology $\mathcal{L}$ consists of board positions, piece codes, files, ranks, and predicates for movement patterns, legality, check, and checkmate.

Unlike the other three domains, chess has a single fixed theory T : the rules of chess. There are no competing doctrines, no soft constraints, no axiom subsets to choose from. Every Skolem witness (every legal position, every valid move sequence) must satisfy all of T . The constraint type is **hard**: a rook cannot move diagonally, period.

The divergence in chess appears not between theories but between **proof strategies**. Given a mate-in-one puzzle (White: Ka6, Rd1; Black: Ka8; solution: Rd8#), the move itself is uniquely determined. But the **verification** — the proof that Rd8 is checkmate — admits multiple strategies that differ in which intermediate lemmas they supply.

We discovered that Z3’s E-matching engine cannot verify the checkmate from raw axioms alone. It requires **proof scaffolding**: explicit Skolem witnesses for the existential quantifiers in `attacks_square`, explicit `in_check` lemmas for each escape square, and explicit king-tracking through nested `apply_move` calls. Different scaffolding strategies are different proof theories over the same position, and they diverge on which lemmas are needed:

$$\Delta(\text{scaffold_1}, \text{scaffold_2}) = \{\text{which intermediate lemmas are supplied}\}$$

The selector here is the prover — human or machine. The choice is strategic: which lemmas flatten the quantifier chains most efficiently?

Chess occupies a unique position in our taxonomy: the object-level theory is fixed (hard constraints), but the meta-level proof theory admits selection. This mirrors the relationship between a formal system and its metatheory in mathematical logic.

4 Hard and Soft Constraints

The four domains reveal a fundamental distinction in constraint type that determines where theory selection operates.

Hard constraints are inviolable. In chess, an illegal move is not a move. In ZFC, an axiom is not optional. Every model must satisfy every axiom. Theory selection, when it occurs, operates at the level of **extensions** (ZFC+CH vs. ZFC+¬CH) or **proof strategies** (which scaffolding to supply), not at the level of axiom subsets.

Soft constraints are violable by choice. In music, a composer can write parallel fifths. The result is not an illegal score — it is a score satisfying a different sub-theory. Theory selection operates directly at the axiom level: the composer chooses $T' \subseteq T$.

This distinction determines the structure of the Skolem space:

- Under hard constraints, the Skolem space is $\text{SAT}(T)$ — all models of the full theory. The freedom is in choosing **which model** (level 2 only).
- Under soft constraints, the Skolem space is $\bigcup_{T' \subseteq T} \text{SAT}(T')$ — the union over all sub-theories. The freedom is in choosing both **which theory** (level 1) and **which model** of that theory (level 2).

Domain	Constraint type	Level 1 (theory)	Level 2 (witness)
Chess	Hard	Fixed	Strategy: which move
Set theory	Hard + extensions	Which extension of ZFC	Which model
Law	Hard base + doctrine	Which doctrine to invoke	Which verdict
Music	Soft	Which axioms to obey	Which notes to write

Table 1: Two levels of selection across domains. Hard constraints fix level 1; soft constraints leave both levels open.

International law occupies a middle position: the base law (Article 2(4)) is hard, but the doctrinal overlay is a choice. The anticipatory doctrine does not violate the base law — it extends it with an additional exception. This is structurally parallel to ZFC + CH: the base axioms are fixed, but the extension is a choice.

The musical case is the most radical: **every** axiom is potentially optional. The space of theories is not a pair (as in law or set theory) but the full power set of the axiom collection. Style is a point in 2^T , and composition is a point in $\text{SAT}(T')$ for the chosen T' . The exponential size of the theory space is what makes music feel like creation rather than discovery.

5 Theory Selection as an Extra-Formal Act

In every domain, the formal system can compute the divergence kernel — identify precisely where two theories disagree — but cannot determine which theory to select. The selection depends on considerations that are outside the formal language $\mathcal{L}$:

- **Set theory:** The choice between ZFC+CH and ZFC+¬CH depends on foundational commitments — set-theoretic pluralism vs. realism, the role of large cardinals, the intended interpretation of the power set. These are philosophical positions, not theorems.
- **International law:** The choice between strict and anticipatory doctrine depends on geopolitical context, institutional authority, evidentiary standards, and political will. The formal system cannot adjudicate whether ‘imminent threat’ is genuinely present; that judgment requires intelligence assessment, historical precedent, and political accountability.
- **Music:** The choice of which axioms to satisfy depends on the composer’s aesthetic intent, the musical tradition being engaged, the expressive goals of the specific passage, and the holistic architecture of the work. No formal theory of tonal harmony can explain why Beethoven writes parallel fifths at measure 13 and nowhere else.
- **Chess:** The choice of proof scaffolding depends on the prover’s understanding of the solver’s limitations, the structure of the quantifier chains, and the strategic insight to identify which intermediate results are needed.

In each case, the formal system provides the **vocabulary of disagreement** (the divergence kernel) and the **verification of consequences** (SAT/UNSAT verdicts under each theory). It cannot provide the **reason for selection**. The reason lives in a different layer: philosophy, politics, aesthetics, or strategy.

This is not a failure of formalization. It is a structural feature. The formal system’s contribution is precisely to make the boundary between the computable and the uncomputeable **visible**. Before computing the divergence kernel, one might not know where the theories disagree. After computing it, one knows exactly which predicates carry the weight of the choice. The system does not make the choice easier, but it makes the choice **precise**.

We can state this as a general principle:

Divergence Kernel Thesis. For any two theories T_1, T_2 over a shared ontology $\mathcal{L}$, the divergence kernel $\Delta(T_1, T_2)$ is formally computable, but the choice between T_1 and T_2 is not derivable from any theory in $\mathcal{L}$.

The thesis is not a theorem — it is a structural observation supported by four independent instances. A counterexample would be a domain where a meta-theory in $\mathcal{L}$ determines the correct theory choice. Such meta-theories exist (e.g., ‘the doctrine consistent with more case law’), but they merely push the selection problem one level up: now one must choose the meta-theory.

6 Computational Demonstration

All four domains are formalized in the Kleis verification language and verified by the Z3 SMT solver. This is not incidental — having all domains on a single platform is essential to the structural comparison. Domain-specific formalizations might introduce domain-specific artifacts; a shared substrate rules this out.

The key architectural feature enabling multi-theory comparison is **type-parameterized axiom isolation**. In the legal formalization, operations like `lawful_self_defense` take a `Doctrine` parameter:

```
operation lawful_self_defense : Doctrine × Act → Bool
```

Each doctrine’s axioms are pinned to its `Doctrine` value (`Strict` or `Anticipatory`), so they coexist without collision. The same object — a specific military act — can be evaluated under both doctrines in a single Z3 context. The solver handles the theory selection as a parameter, and verdicts are compared directly.

In the music formalization, the separation is structural rather than parametric: the seven axiom-checking functions are independent, and each returns a verdict. A sub-theory T' is specified by choosing which functions to require SAT. The divergence kernel between Bach-style and Beethoven-style is identified by running both verdict profiles and diffing.

In set theory, the two theories are separate Kleis structures loaded into the same Z3 context. The projection map Π is a Kleis function, and fiber construction is verified by Z3 in under 30 seconds.

In chess, the proof scaffolding is encoded as additional axioms in a `Puzzle` structure. Different scaffolding strategies are different structures with different lemma sets, all referencing the same base `Chess` structure.

The shared platform makes the following operations possible:

1. **Cross-domain comparison:** the divergence kernel has the same structure (small, identifiable, with a pivot predicate) in all four domains.
2. **Constraint-type classification:** hard vs. soft is visible in the formalization — chess axioms have no `Doctrine` parameter; music axioms are independent functions.
3. **Reproducibility:** all results are machine-checked from source files. No domain requires manual reasoning steps.

7 Discussion

Relationship to model theory. The divergence kernel is a computational instantiation of a well-known model-theoretic phenomenon: theories that agree on a fragment of the language may disagree on extensions. The contribution is not the observation itself but its computational demonstration across heterogeneous domains on a single platform, and the identification of pivot predicates that localize the disagreement.

Relationship to independence proofs. In our companion paper on projection fibers, we showed that logical independence is non-invariance on the fibers of a projection map. The divergence kernel complements this: where projection fibers detect independence **within** a theory (CH is independent of ZFC), the divergence kernel characterizes disagreement **between** theories (ZFC+CH vs. ZFC+¬CH). The two notions are dual: independence generates the theories whose divergence kernel the new framework computes.

Soft constraints and the power-set problem. When constraints are soft, the space of sub-theories is $2^{|T|}$. For our seven-axiom TonalHarmony theory, this gives $2^7 = 128$ sub-theories. Each sub-theory defines a distinct style profile. The combinatorial explosion is the formal content of the intuition that music is ‘more creative’ than chess: the theory space is exponentially larger. Whether this exponential scales to richer theories (hundreds of axioms) is an open question.

Generative AI and the selection boundary. Our framework makes precise a claim that is often stated vaguely: that generative AI ‘lacks understanding’ or ‘merely imitates.’ In our terms, a generative model samples from a distribution over the witness space of an implicit theory. It does not select a theory; it does not compute a divergence kernel; it does not know where the theories disagree. A composer, jurist, or mathematician operates at both levels: selecting the theory and constructing a witness. The formal system makes the boundary between these levels visible.

Limitations. The divergence kernel as defined requires decidable theories (for the SMT solver to return verdicts). Undecidable theories may have kernels that are not effectively computable. Our four domains are all within the decidable fragment of first-order logic used by Z3. Extending the framework to higher-order logic or undecidable arithmetic would require approximate methods (bounded model checking, abstraction).

We also note that identifying the **minimal** generator of the kernel is in general harder than computing individual membership. Our domains have small enough kernels that exhaustive querying suffices, but scaling to theories with hundreds of axioms would require heuristic search.

8 Conclusion

We have defined the divergence kernel $\Delta(T_1, T_2)$ as the set of propositions on which two theories over a shared ontology disagree, and demonstrated across four domains that this set is small, identifiable, and corresponds to a recognizable domain concept: the Continuum Hypothesis, the imminent-attack predicate, parallel-fifth tolerance, and proof scaffolding.

The pattern is invariant: in every domain, the formal system can compute the locus of disagreement but cannot determine which side to inhabit. Theory selection is an extra-formal act — foundational, political, aesthetic, or strategic — that operates outside the language in which the theories are expressed.

We distinguish hard constraints (inviolable axioms, as in chess and base set theory) from soft constraints (selectively violable axioms, as in music), and show that this distinction determines whether theory selection operates at the extension level or the axiom-subset level. The musical case is extreme: every axiom is optional, and style is a point in the power set of the theory.

All four domains run on the same verification substrate (Kleis + Z3), enabling direct structural comparison and ruling out domain-specific artifacts. The divergence kernel is not a metaphor; it is a computable object that localizes disagreement to specific, identifiable predicates.

The deepest implication is perhaps this: in every domain where formal reasoning applies, there is a boundary between what the system can determine and what the human must choose. The divergence kernel makes this boundary precise. It does not resolve the choice — that is the mathematician’s commitment, the jurist’s doctrine, the composer’s art — but it reveals exactly where the choice lives, and exactly what depends on it.

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Appendix

A Divergence Kernel Examples

Set theory. Let T_{base} be the cardinal-arithmetic fragment of ZFC (ordering, addition, multiplication, exponentiation, power-set growth law). Let $T_1 = T_{\text{base}} \cup \{\mathfrak{c} = \aleph_1\}$ and $T_2 = T_{\text{base}} \cup \{\mathfrak{c} > \aleph_1\}$. Then:

$$\Delta(T_1, T_2) = \{\varphi \in \text{Sent}(\mathcal{L}) \mid \varphi \text{ depends on the value of } \mathfrak{c}\}$$

The generator is the single sentence $\mathfrak{c} = \aleph_1$.

International law. Let T_{base} consist of Article 2(4), necessity, proportionality, and Security Council authorization. Let $T_1 = T_{\text{base}} \cup \text{StrictArticle51}$ and $T_2 = T_{\text{base}} \cup \text{AnticipatoryArticle51}$. The only structural difference is the disjunct $\vee \text{imminent_attack}(\dots)$ in T_2 's self-defense precondition. The generator is:

$$\text{imminent_attack}(\text{target}(a), \text{attacker}(a)) \text{ being satisfiable for some } a$$

Music. Let T be the seven-axiom TonalHarmony theory. Let $T_{\text{Bach}} = T$ (all seven axioms enforced) and $T_{\text{Beethoven}} = \{\text{axioms } 1, 2, 4, 5\}$ (tonal cohesion only). The generators of $\Delta(T_{\text{Bach}}, T_{\text{Beethoven}})$ are:

$$\{\text{axiom 3 (triadic arpeggiation), axiom 6 (no parallel fifths), axiom 7 (harmonic rhythm)}\}$$

Chess. Let T be the fixed rules of chess. There is no theory-level divergence: $\Delta(T, T) = \emptyset$. Divergence appears at the proof-strategy level, where different scaffolding structures supply different intermediate lemmas to guide the SMT solver.

B Kleis Source Files

All formalizations referenced in this paper are available as executable Kleis source files:

- **Set theory:** `examples/cantor/cantor_set_theory.kleis` (shadow theory, 19 examples), `examples/cantor/projection_fibers_paper.kleis` (full paper, 46 verified results)
- **International law:** `examples/authorization/un_charter_article51.kleis` (7 verified theorems, doctrine comparison)
- **Music:** `stdlib/theories/tonal_harmony.kleis` (7 axiom checkers, pitch arithmetic), `examples/music/moonlight_sonata.kleis` (14-measure AST), `examples/music/moonlight_paper.kleis` (full paper, 12 verified results)
- **Chess:** `examples/chess/chess_mate_in_two.kleis` (back-rank mate, 9 verified assertions with proof scaffolding)

All files are self-contained and can be verified with `kleis test <filename>`.